

Latex-Vorlage für Abschlussarbeiten und Projektarbeiten mit sehr sehr sehr langem Titel um die Formatierung zu testen

Scientific work to obtain the degree

Master of Science (M.Sc.)

at the TUM School of Engineering and Design
of the Technical University of Munich.

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Abstract

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Zusammenfassung

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Acronyms

BGU TUM School of Engineering and Design

BIM Building Information Modeling (STEIN, 2014)

List of Symbols

g	$\left[\frac{m}{s^2}\right]$	Gravity of Earth
R_A	$[m]$	Earth radius

Chapter 1

Introduction

$\LaTeX$ is a text typesetting system that can be used to create larger pieces of writing, such as books, theses, master's theses and articles. Since Latex offers excellent functionalities for creating bibliographies, indexes, and glossaries, it is an ideal choice for creating such a scientific work. (STEIN, 2014)

This template is intended to facilitate the creation of theses at TUM School of Engineering and Design (BGU). To ensure the best possible clarity with as much functionality as possible, a separate class was created for this purpose. To use it, the files BGUthesis.cls as well as BGUngerman.def and BGUenglish.def must be present in the main directory next to the thesis.tex file.

Important packages for frequently used functions are integrated directly in the class and do not have to be included separately, a list with the incorporated packages is attached [appendix A](#). In the following, some commands and mechanisms are explained or used and should be comprehensible by the sample code underlying this document. If you are unsure, it is helpful to do some research on the web. The following websites are worth mentioning for general information and instructions on $\LaTeX$ commands and the included packages:

- <http://de.wikipedia.org/wiki/LaTeX>
- <http://www.ctan.org/>
- <http://www.golatex.de>

1.1 Software

To make use of $\LaTeX$, a number of different $\LaTeX$ editors as well as different $\LaTeX$ distributions can be used:

- <http://www.miktex.org/>
MiKTeX is a TeX distribution for Microsoft's Windows operating systems. A simple TeX editor (TeXworks) is also supplied with the installation package.
- <http://blog.kowalczyk.info/software/sumatrapdf/free-pdf-reader-de.html>
Sumatra PDF is a lightweight, free, open-source PDF, ePub, MOBI, XPS, DjVu, CHM, CBZ, CBR viewer for Windows. Unlike Arcobat Reader, PDF files opened in Sumatra can still be modified by other programs.

- <http://www.texniccenter.org/>

TeXnicCenter (TxC) is a free text editor for LaTeX documents under Windows.

- <https://www.xm1math.net/texmaker/>

TeXmaker is a freely available cross-platform (Windows, MacOS, Linux) LaTeX editor.

Alternatively, online solutions such as „Overleaf“ can be used. TUM also offers its own Overleaf server (<https://latex.tum.de>). With these online tools, a $\text{\LaTeX}$ project can be created and edited directly in the browser without having to install the aforementioned software on the computer. In addition, other people can be invited to the project, who can then edit the documents at the same time.

1.2 First steps

Before you can start writing the scientific paper, you should first adjust the settings in the header of the file `thesis.tex` (cf. Algorithm 1.1). Lines 2–8 influence the formatting and some predefined macros. Especially important are lines 6–8, which set the language, the citation style and the document type. The most important options are explained below:

line 2: Aligns equations (`equation-environment`) flushed left

line 3: Sets the font size of the body text (`\normalsize`). Other font sizes will be aligned accordingly.

line 4: Specifies whether the document is single-sided (`oneside`) or double-sided (`twoside`).

line 5: If `print` is set to `true`, all links, references, and code sequences in the text are no longer colored. This makes the document have fewer color pages and cheaper to print.

Algorithm 1.1: class options

```

2 fleqn, % left aligned equations.
3 11pt, % normal textsize.
4 oneside, % single side page, use twoside if >50 pages.
5 print=false, % sets all link colors to black (lower printing costs).
6 language=english, % use only ngerman or english! the other language is
  loaded as secondary language.
7 citeStyle=apa, % citation style, alternative: iee or none if using own
  style
8 type=thesis, % thesis for TUM corporate design title and declaration,
  alternative: diss for dissertation title and declaration, plain for
  custom titlepage, dissOld for old (not TUM Corporate Design)
  pagestyle

```

- line 6:** Sets in which language the document is written. Only English and German should be used (`ngerman` and `english` respectively). If you want to use another language, you have to create a translation of the `.def` file.
- line 7:** The citation style to be used should be entered here. So far only `apa` and `ieee` are predefined. If another style is to be used, please enter `none` here and include and set up the `biblatex` package independently (cf.).
- line 8:** Specifies the type of the paper. The options here are `thesis`, for a bachelor or master thesis, `diss`, for a dissertation, or `plain`, for a free-form title page. To customize the `plain` title page, simply renew the `\TPplain` command.

other class options:

Option	Type	Function
<code>colorcode</code>	<code>true/false</code>	If set to <code>false</code> , keywords in <code>listing</code> -environments will not be printed in color. Has no effect, if <code>print=true</code> .
<code>microtype</code>	<code>true/false</code>	If there are font expansion errors, this option should be set to <code>false</code> . Otherwise scalable Fonts need to be installed for $\text{\LaTeX}$ (more difficult).
<code>showframe</code>	<code>true/false</code>	Displays, if set to <code>true</code> , the borders of any content box. Makes it easier to debug any overfull hbox issues.
<code>citeInAcroList</code>	<code>true/false</code>	Determines if references should be included in the list of acronymes. [Default <code>false</code>]
<code>chairLogo</code>	<i>string</i>	If a chair logo should be printed in the title page header, specify the filename here and place the image in the <code>figures/Logos</code> folder.

Next, all commands in the **lines 12–35 must** be adapted to your own needs. To do this, simply adjust the text in the last bracket of the respective `Set...` command. These commands are used in the predefined macros, `\maketitle` and `\Declaration`, which format the personalized title page at the beginning and affidavit at the end.

1.3 Packages included in CMStthesis class

The following packages are loaded directly from the CMStthesis class and are ready for use:

Package name	Description
listings:	Needed to properly typeset programming code.
tikz	Used for creating vector graphics.
rotating	Necessary to rotate float objects (tables or images).
tabularray	For creating tables. Comprehensive library that combines the functionality of some other table packages (e.g. longtable, tabularx, multirow, etc.).
caption	For fine-tuning table and image captions.
floatrow	Provides better control over the arrangement of float objects (tables, images).
acro	Used for abbreviations and the corresponding list of abbreviations and the list of symbols.
cleveref	Simplifies and improves referencing. Instead of <code>\ref</code> <code>\cref</code> should be used.

Other required packages (they are already loaded, please note to avoid package incompatibilities):

- | | |
|----------------------------------|-------------|
| - iftex | - booktabs |
| - etoolbox | - footnote |
| - kvoptions | - subfig |
| - xstring | - biblatex |
| - xcolor | - titlesec |
| - graphicx | - fancyhdr |
| - enumitem | - hyperref |
| - amsmath,amssymb,amsthm,nccmath | - showframe |
| - framed | - microtype |
| - url | - csquotes |
| - pgfplots | - lscap |

1.4 Chapters and Sections

To structure the work into chapters, sections and subsections, the commands corresponding to [Algorithm 1.2](#) are used. From these, a table of contents is automatically created using the command `\tableofcontents` (see `thesis.tex`, line 92). The `\label{}` and `\cref{}` commands can be used to refer to a chapter or paragraph in the text. The

commands for chapters, paragraphs, etc. exist in the version shown, as well as with an appended asterisk, e.g. `\chapter*{}`. The star version creates unnumbered chapters, sections, etc., which are not listed in the table of contents, e.g. to further divide the document into more sections without cluttering the table of contents.

Algorithm 1.2: Chapters and Sections

```
\chapter{Chapter 1}
\label{ch:kapitel_1}

\section{Section 1.1}
\label{sec:abschnitt_1.1}

\subsection{Section 1.1.1}
\label{sec:abschnitt_1.1.1}

\subsubsection{Section 1.1.1.1}
```

If special characters or other $\text{\LaTeX}$ commands are to be used in a heading, such as abbreviations, then it is recommended to convert these characters to a string using the `\texorpdfstring` command. Otherwise, the `hyperref` package may cause an error with it. For example, a heading might contain the characters, $\mathcal{L}(\gamma)$. The math environment and special characters are problematic here. In the heading this should be realized with `\texorpdfstring{\mathcal{L}(\gamma)}{Lg}`.

To keep the data structure of the $\text{\LaTeX}$ project clean it is useful to create a separate `.tex` file for each chapter and include it in the main document (`thesis.tex`) with the command `\include{<filename>}`, cf. lines 111–113 in `thesis.tex`. This way, any errors can be better localized and clarity is improved when editing the thesis.

1.5 Colors

[table 1.1](#) displays the colors predefined in the document class, which can be switched off with the switches `print` and `colorcode` (cf. [Algorithm 1.1](#), line 5 and 7) „can be switched off“ (see description in [section 1.2](#)). By „off“ is meant here that the color is then replaced by a gray value to e.g. create a print version of the document (as a test this document can be compiled with the option `print=true`).

The colors can be changed with `\DefineColor[<gray value>]{<name>}{<model>}{<color>}` (watch out for capital letters!) can be overwritten or recreated. If *mix* is used as `<model>`, the color is defined with the `\colorlet` command, otherwise with `\definecolor` (cf. [xcolor documentation](#)). The optional gray value allows to specify a grayscale value if the color „is turned off“.

Examples:

```
\DefineColor[0.2]{beispiel1}{RGB}{22,0,250}
```

Table 1.1: Predefined colors.

<name>	<model>	<color>	<gray value>
<code>kwcolor</code>	mix	blue!70!black	0.1
<code>kwcolor2</code>	mix	magenta!80!white	0.8
<code>kwcolor3</code>	mix	OliveGreen	0.3
<code>kwcolor4</code>	mix	Yellow!50!white	0.6
<code>kwcolor5</code>	mix	blue!50!white	0.7
<code>kwcolor6</code>	mix	RedViolet	0.4
<code>commentcolor</code>	mix	green!50!black	0.4
<code>commentcolor2</code>	mix	gray	0.5
<code>stringcolor</code>	mix	violet!50!black	0.1

```
\DefineColor[0.2]{beispiel2}{mix}{Blue!60!White}
```

1.6 Images

Images are best included in the `figure` float environment with `\includegraphics` (see [Algorithm 1.3](#)). This command supports most pixel graphics formats as well as formats, such as EPS and PDF, for vector graphics. If you want to include Inkscape files (`.svg`) directly, you should use the `svg` package (not included in the document class, cf. [svg documentation](#)). This allows SVG files to be included in the document with the `\includesvg` command.

If Matlab plots are to be displayed in the document, the `matlab2tikz` project is recommended (cf. <https://github.com/matlab2tikz/matlab2tikz>). With this Matlab tool plots can be converted into a TikZ representation (vector graphics tool for $\text{\LaTeX}$), which can be embedded directly in the document.

Algorithm 1.3: Embedding image files.

```
\begin{figure}
  \centering
  \includegraphics{<filename>}
  \caption{My image description}
  \label{fig:my_label}
\end{figure}
```

For better formatting freedom for images, the `floatrow` package has been integrated into the document class (see [floatrow documentation](#)). This package is very good for displaying images next to and below each other. For example, if you need to display images with the same height next to each other, the package provides a very simple solution (cf. [Algorithm 1.4](#)). Without the `\ffigbox` command, if multiple images are

included in a `figure` environment, each with its own caption, the `\RawFloats` command must be executed at the beginning of the environment.

Algorithm 1.4: Display of two images side by side with the same height and each with its own caption.

```

\begin{figure}
  \CommonHeightRow[5.5cm]{%
    \begin{floatrow}[2]
      \ffigbox[\FBwidth]{%
        \caption{Bildunterschrift 1.}%
        \label{fig:my_label_1}%
      }{%
        \includegraphics[height=\CommonHeight]{<filename>}%
      }
      \ffigbox[\FBwidth]{%
        \caption{Bildunterschrift 2.}%
        \label{fig:my_label_2}%
      }{%
        \includegraphics[height=\CommonHeight]{<filename>}%
      }
    \end{floatrow}
  }
\end{figure}

```

A list of figures is created automatically with the command `\listoffigures` (see `thesis.tex`). In the directory, the figure number and the corresponding caption (`\caption`) are displayed. If the caption is too long, the corresponding `caption` command can be used to specify an alternative, shorter caption in square brackets, which will only be displayed in the directory.

1.7 Code

Code can be easily and conveniently displayed using the listings package. Within the `lstlisting` environment, all spaces are considered and all commands are ignored unless a command is escaped. In this template, commands inside the `(*)` and `(*)` strings are escaped, i.e. executed as `LATEX`code. This way labels can be attached to important lines and referenced accordingly, e.g. [line 5](#) in the [Algorithm 1.5](#). However, this only works if line numbering has been enabled via the appropriate environment option.

Templates for three programming languages have already been defined in `thesis.tex`, which can be used to format keywords defined within the `lstlisting` environment according to the template. To use the appropriate language, it can either be set globally with `lstset`, or selected in each case in the corresponding `lstlisting` environment via the `language` option.

Without further ado, the `lstlisting` environment is not a float environment, like e.g. `figure` or `table`, and is always placed in the body text. However, a caption, label and position specifier (h,t,b) can also be defined here via appropriate options.

Algorithm 1.5: Example listing

```
1 This is how
2   code looks like
3 with the listings package
4
5 In this line a label is defined and can be referenced.
```

1.8 Abbreviations

To create abbreviations please follow the instructions in the *00c_abbreviations.tex* file (see also [acro documentation](#)). In general, it is recommended to create an entry for each abbreviation and to use one of the `\ac` commands each time it is used in the text. This way, each abbreviation is linked to the list of abbreviations via the `hyperref` package (cf. [table 1.2](#)).

Table 1.2: abbreviations, examples

acf (repeated 1st use):	Building Information Modeling (BIM) (STEIN, 2014)
acl and acs in the sentence:	Building Information Modeling, abbreviated as BIM .
ac (after first use):	BIM
accite:	(STEIN, 2014)

1.9 List of Symbols

Table 1.3: Symbole, Beispiele.

Erdbeschleunigung:	g
Erdradius:	R_A

1.10 Citations and Bibliography

In order to properly mark citations and create a bibliography automatically, the individual sources must be exported to a `.bib` file and included in `thesis.tex` with the command `\addbibresource`. For ease of reference, `.bib` files should be placed in the separate `literature` folder. These `.bib` files can also be edited by hand, as a template the existing `references.bib` file can be used as a template. The Bibtex code of publications can be obtained for example from Google Scholar (<http://scholar.google.de/>). If Citavi is used for literature management, a Bibtex file can also be exported directly there.

Each entry in the `.bib` file must have its own unique Bibtex key, with the help of which the respective source can be cited in the work. If the APA citation style is used as recommended, there are two different $\text{\LaTeX}$ commands that can be used for citation.

First, there is the `\parencite` command, which can be used to set the citation in the bracketed style. This type of quotation is usually placed at the end of the sentence (before the period) or the part of the sentence to which the quotation refers. Alternatively, the citation can be placed at the end of a paragraph (then after the period), provided that the source refers to the entire paragraph.

On the other hand, there is the `\textcite` command which outputs the reference without parentheses. This type of citation is used when the author of the source is directly mentioned by name in the text. For both commands there is also the option to define a prefix and/or postfix, e.g. if the page number is to be given or if e.g. a „she“ is to be prefixed.

Example citation, in parentheses: (AMANN et al., 2013)

Continuous text citation: STEIN (2014) gives many good references in his $\text{\LaTeX}$ introduction.

Bracketed citation with prefix: (siehe auch AMANN et al., 2013)

Bracketed quote with postfix: (AMANN et al., 2013, S. 5f.)

If not APA but IEEE is used, just `\cite` can be used, as there are no different versions here.

Troubleshooting

The default package for literature management in this template is `biblatex`, which is set with the backend `biber`. In most $\text{\LaTeX}$ editors, an extra setting must be made to use `biblatex` with this setting. The backend `biber` is a separate $\text{\LaTeX}$ program which compiles the auxiliary files necessary for setting the bibliographic references. In order for this program to be executed when compiling the project data, a setting must be made in the corresponding $\text{\LaTeX}$ editor. What has to be done at the editor differs depending on the editor. A comprehensive description for the most common editors can be found at the following address: <https://texwelt.de/fragen/1909/wie-verwende-ich-biber-in-meinem-editor>.

If there are further problems with the default settings, you can also enter `citeStyle=none` in the class options in line 10. With this setting no package is loaded for the literature management, it can be loaded then another package and own attitudes be made. As an alternative to `biblatex` the `natbib` package is recommended, but it is somewhat outdated and has not been updated since 2010. If the problem is with `biber`, it is also possible to load the `biblatex` package with the option `backend=bibtex`, this way no configuration of the editor is necessary, but it is not possible to use the APA citation style this way.

1.11 Miscellaneous

Line breaks can be created with `\\`. However, occasionally this creates a „underfull box“ and it is therefore safer to „fill the line with a `\hfill` before the line break. For the sake

of simplicity, the command `\n1` has been defined for this purpose, which can be used to break a line in continuous text without any problems.

A new paragraph is obtained by inserting a blank line in the code. Examples for this line and paragraph structuring can be found in the code of this description.

Other predefined commands

Table 1.4: caption

<code>\dC:</code>	Represents a short form of $\text{\textcircled{C}}$ and writes degree Celsius ($^{\circ}\text{C}$) ^a
<code>\TblrNote</code>	Allows placing table footnotes. They can be referenced multiple times, such as the footnote <i>a</i> in the first and second lines. ^a
<code>\lquote</code>	Creates a quotation, i.e. an indented area in quotation marks and with source citation (see examples in chapter 2).
<code>\lquote</code>	Creates a quote similar to <code>\lquote</code> only in body text and not indented.
<code>\Declaration</code>	Creates the affidavit that must be signed for the work to be validly submitted. Is already called in <code>thesis.tex</code> at the end.

^a Example for `\TblrNote`. Only available in `longtblr` and `talltblr`. Must be added to `longtblr/talltblr` options.

Chapter 2

Lorem Ipsum

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2.1 Und so weiter...

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2.2 ...und so fort...

At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Appendix A

Bezeichnung des Anhangs

Liste der verwendeten Pakete...

Hier kann ein Anhang eingefügt werden.

Bibliography

- AMANN, J., BORRMANN, A., HEGEMANN, F., JUBIERRE, J. R., FLURL, M., KOCH, C., & KÖNIG, M. (2013). A refined product model for shield tunnels based on a generalized approach for alignment representation. *Proc. of the ICCBEI 2013*.
- STEIN, S. (2014). Eine kleine latex (tex) einföhrung [Stand: 2014-10-21].

Declaration

I hereby affirm that I have independently written the thesis submitted by me and have not used any sources or aids other than those indicated.

Location, Date, Signature